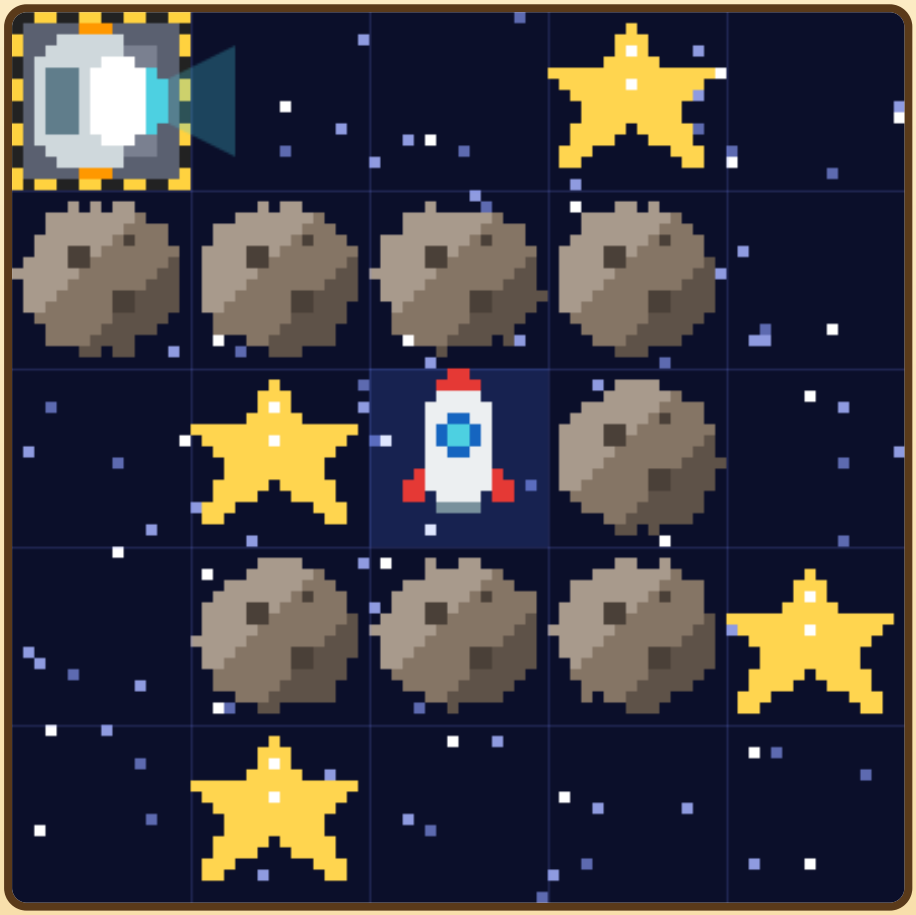
For kids and teachers

# Learn Programming in Space

From blocks to Python — 13 star-hunting stages



National Institute of Technology, Maizuru College — Yasuhito Inoue

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# How to use this book

Use this book together with the “Space Star Hunt” app on a computer or tablet. You will learn how to think like a programmer by building programs with blocks and with Python.

1. Read the **problem** and look at the picture.
2. Think about the order: trace it with your finger or write it in your notebook.
3. Build the program in the app and press “▶ Run”. If it doesn’t work, find what was different and fix it.
4. Stuck? Look at the 💡 **Hint**. When you are done, turn the page and check the 📖 **Answer** and the ✨ **Key point**.

There is more than one right answer. If it works, your program is correct, even if it is different from the book. Mistakes are hints — try again and again!

## Apps in this book

| App | Tasks | Grade |
|---|---|---|
| Space Star Hunt | 7 | Grade 3+ |
| Python Space | 6 | Grade 5 – junior high |

PART 1

# Part 1 The space book for kids

Chapter 1

# Collect stars and go to the rocket

Move the astronaut, pick up every ⭐ star, then board the 🚀 rocket. You bump into rocks, so plan your path.

| Command | What it does |
|---|---|
| move forward | Moves one square the way it is facing |
| turn right / turn left | Changes direction on the spot (does not move) |
| ⭐ pick up star | Picks up the star on this square (fails if there is none) |
| if there is a ⭐ star here | Runs the commands inside only when there is a star |
| 🚀 repeat until mission complete | Repeats until all stars are collected and you reach the rocket |

## The big idea in this app

**Check first, then act.** Even if you don’t know where the stars are, you can collect them by checking every time. Automatic doors and sensor lights also check before they act.

## When it doesn’t work

- Look where it stopped, and whether any stars are left.
- On the worksheet, write the number of stars collected after each move.

Chapter 2

# 🚀 Space Star Hunt

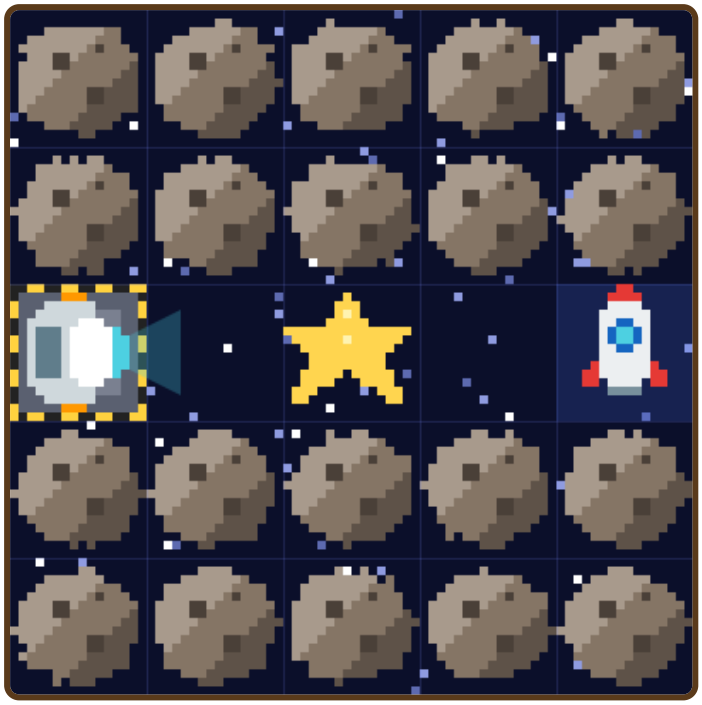
Collect stars and go to the rocket. You will go from **order → loops → patterns → if → any space**. The last stage is tested on 30 random worlds.

**Commands in this chapter**

move forward / turn right, left / ⭐ pick up star / repeat ○ times / if there is a ⭐ star here / if ○ is open / 🚀 repeat until mission complete

# Stage 1: Pick up a star

Sequence



The start

**Problem**

Pick up every star ⭐, then board the rocket 🚀! Use "pick up star" on a star square.

🎯 **Goal**：Use “pick up”

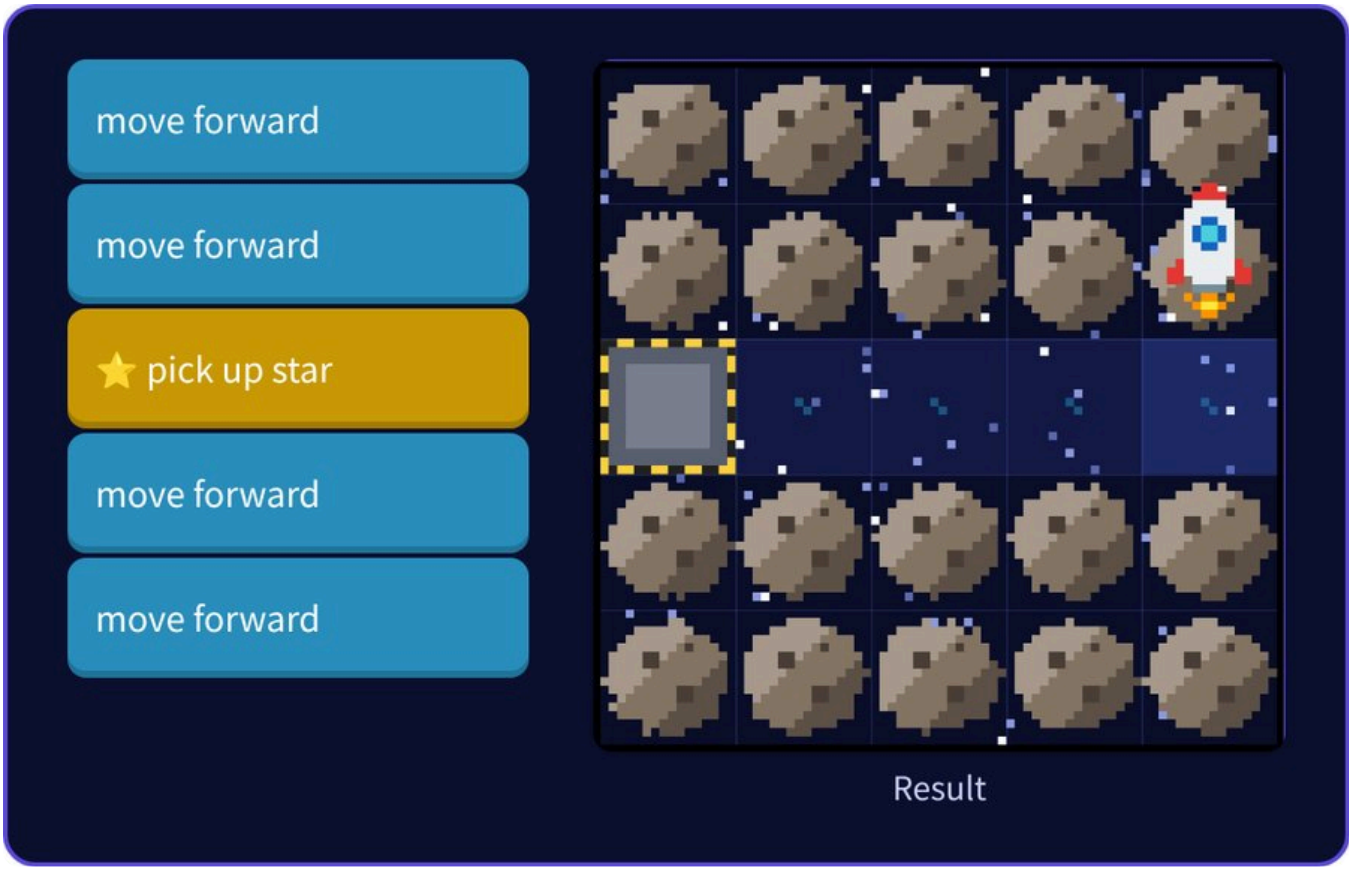
💡 **Hint (look if you get stuck)**

Forward 2 → pick up star → forward 2

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 1: Pick up a star




**✨ Key point**

Use “pick up” on the star square. Collect them all, then go to the rocket!

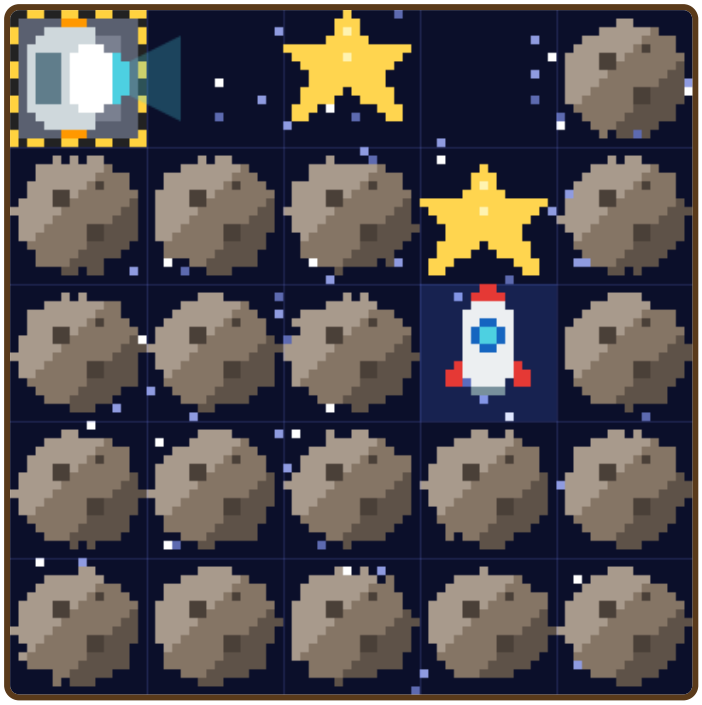
**⚠ Common mistake**

Picking up on a square with no star fails. Count where to pick up.

🎬 Video: `videos/en/space-01.mp4`

# Stage 2: Turn and collect

Sequence Direction



The start

**Problem**

There are 2 stars. Be careful where you turn.

🎯 **Goal**：Turn and collect in order

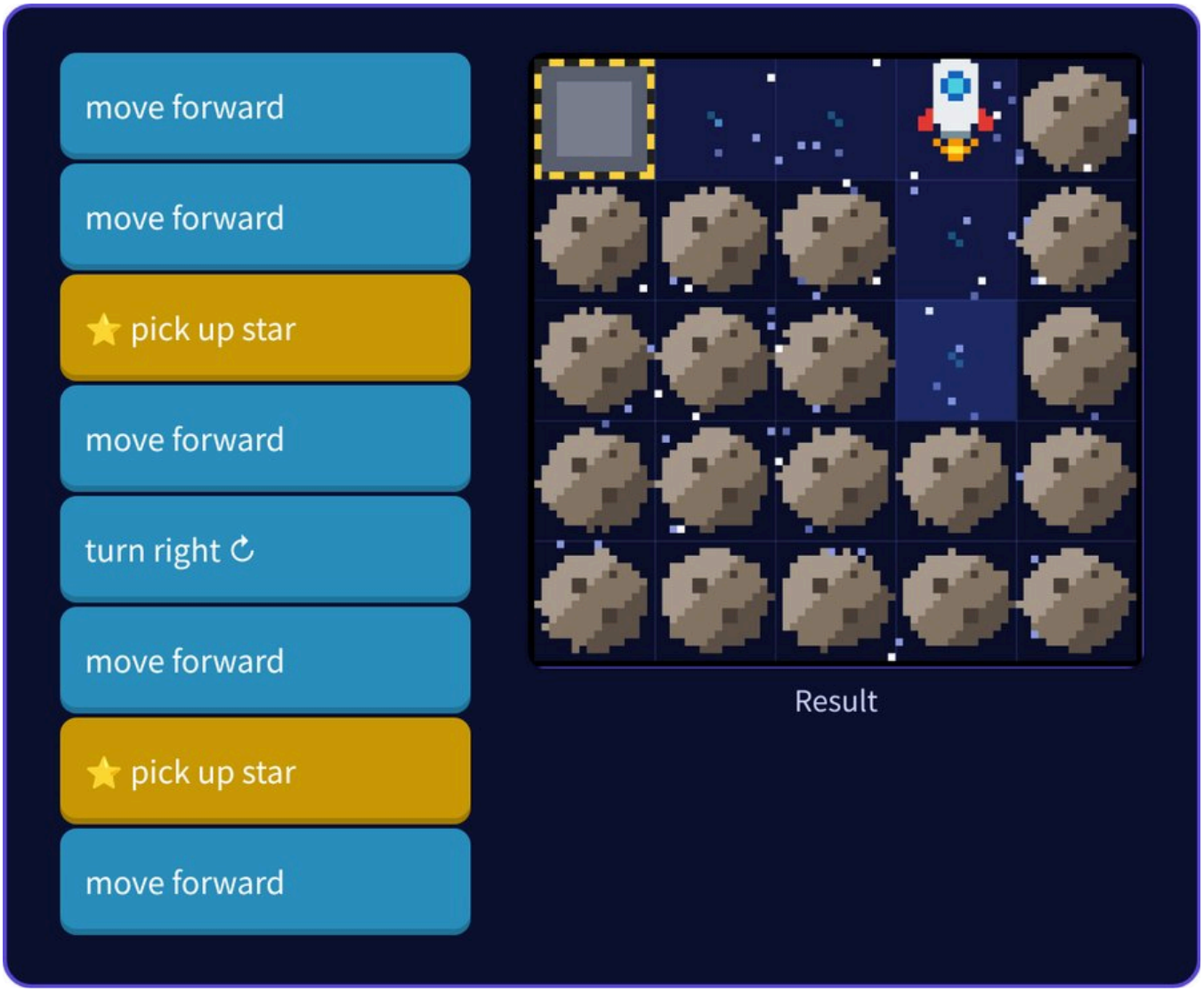
💡 **Hint (look if you get stuck)**

Forward 2 and pick up → forward 1 → turn right → forward 1 and pick up → forward 1

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 2: Turn and collect




✨ **Key point**

The order of turning and picking up is important.

⚠ **Common mistake**

Watch the order: pick up before or after turning?

🎬 Video: `videos/en/space-02.mp4`

# Stage 3: Collect with a loop

Loops



The start

**Problem**

The stars are in a row. Put "move → pick up" into a loop.

🎯 Target: 4 blocks or fewer

🎯 **Goal**：Collect with a loop

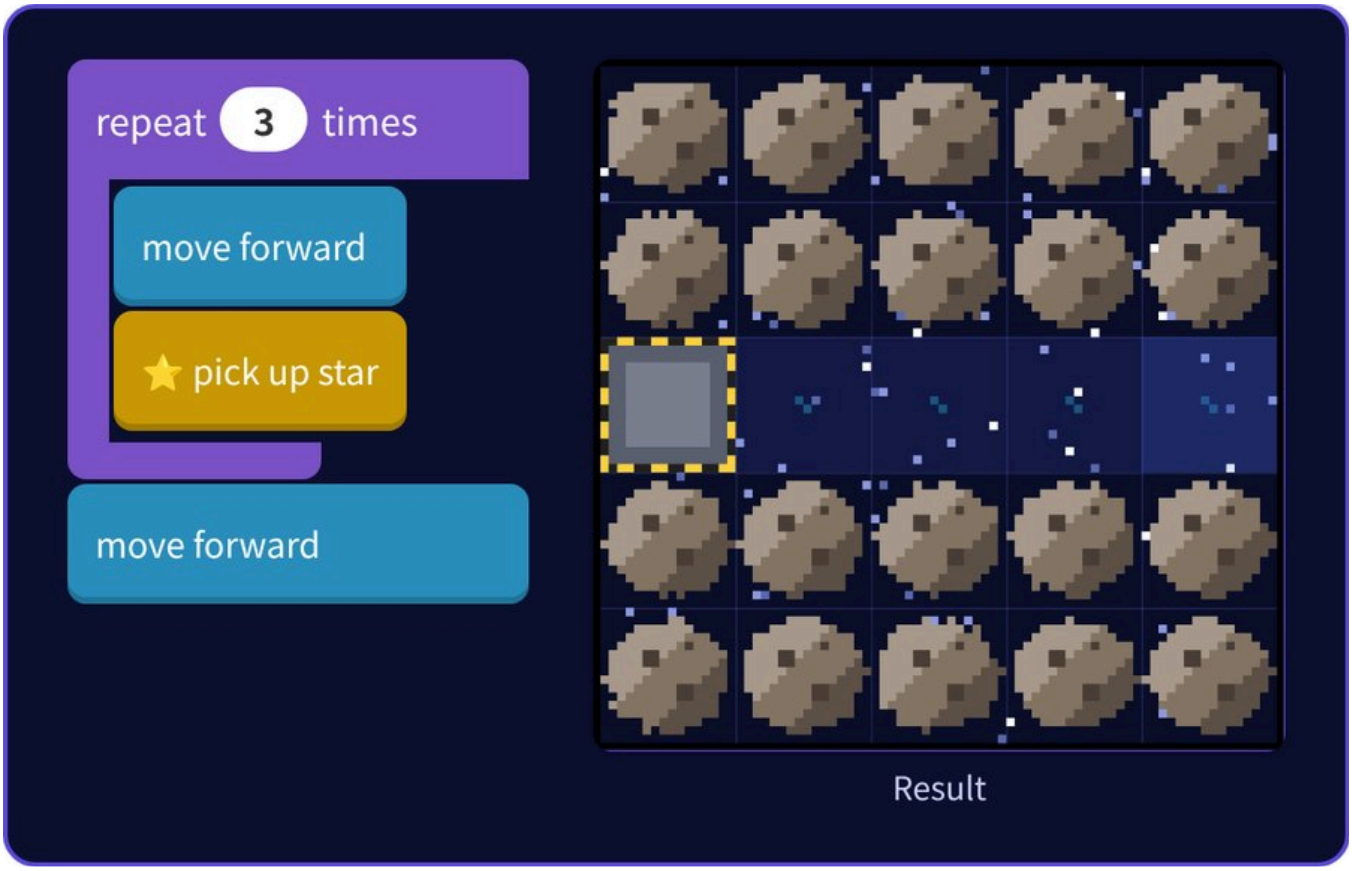
💡 **Hint (look if you get stuck)**

"repeat 3 (move forward → pick up star)" → "move forward"

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 3: Collect with a loop




**✨ Key point**

Repeat “move → pick up” 3 times, then move one more square.

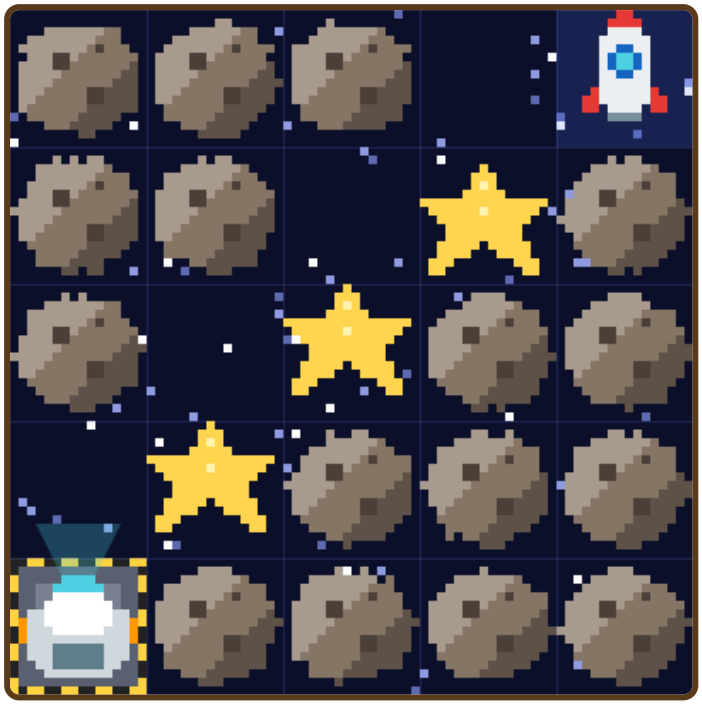
**⚠ Common mistake**

Forget the last “move” and you will not reach the rocket.

🎬 Video: `videos/en/space-03.mp4`

# Stage 4: Star stairs

Loops Patterns



The start

**Problem**

There are stars on the stairs. Find the set of moves that repeats.

🎯 Target: 6 blocks or fewer

🎯 **Goal**：Find the pattern and repeat it

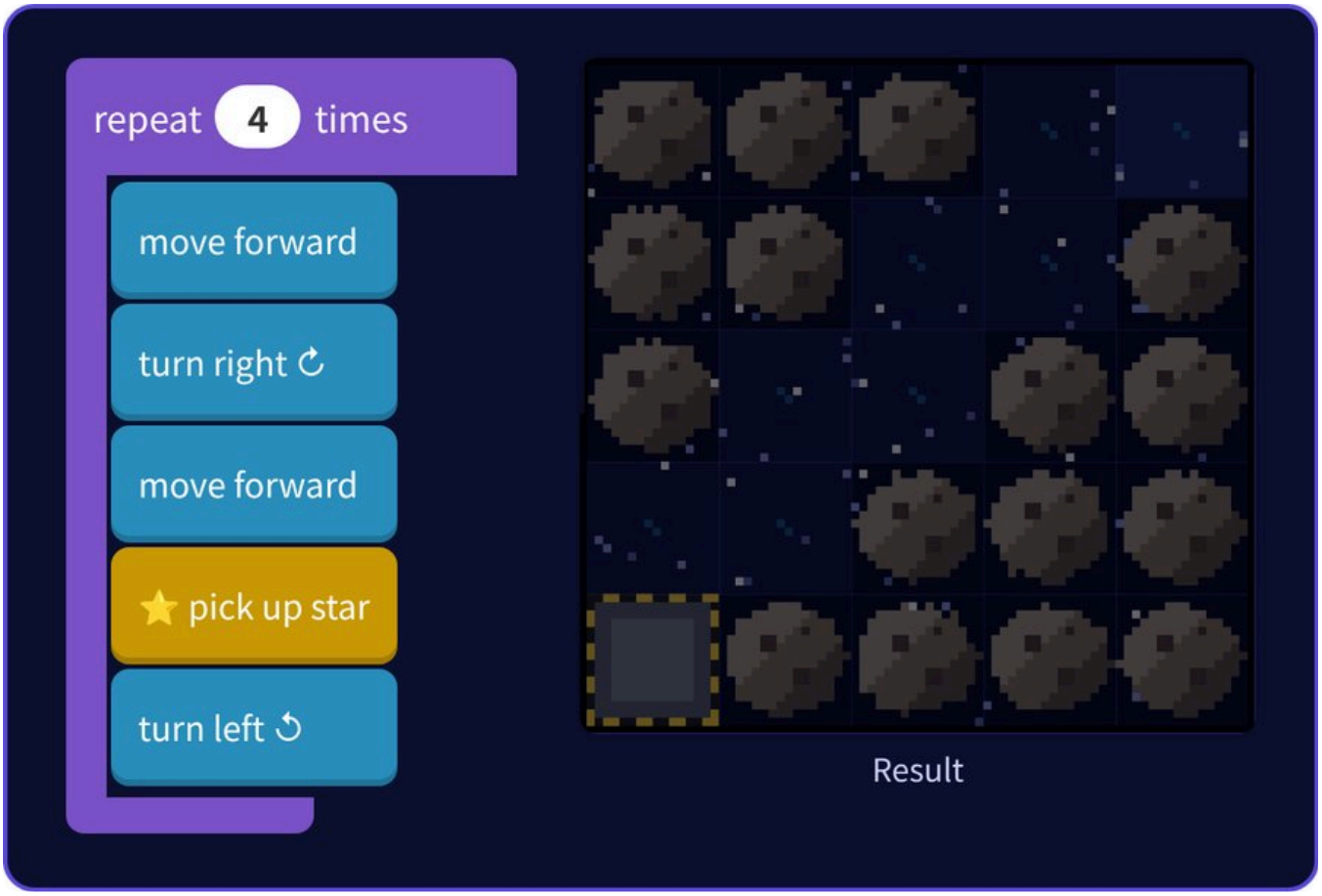
💡 **Hint (look if you get stuck)**

Repeat "forward → turn right → forward → pick up → turn left". You finish when you reach the rocket.

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 4: Star stairs



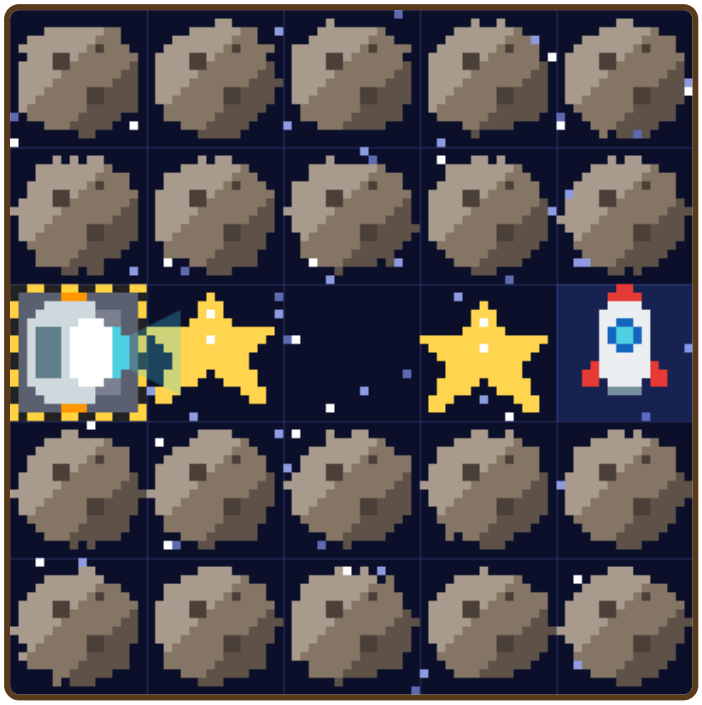

> ✨ **Key point**
>
> Each step of the stairs is one set. You reach the rocket during the 4th repeat.

> ⚠ **Common mistake**
>
> If “pick up” is in the wrong place, you pick up where there is no star.

🎬 Video: `videos/en/space-04.mp4`

# Stage 5: If there is a star

Conditions Loops



The start

**Problem**

Some squares have stars and some don't. Picking up where there is no star is a mistake! Let the explorer check first.

🎯 Target: 4 blocks or fewer

🎯 **Goal**：Check first, then act (if)

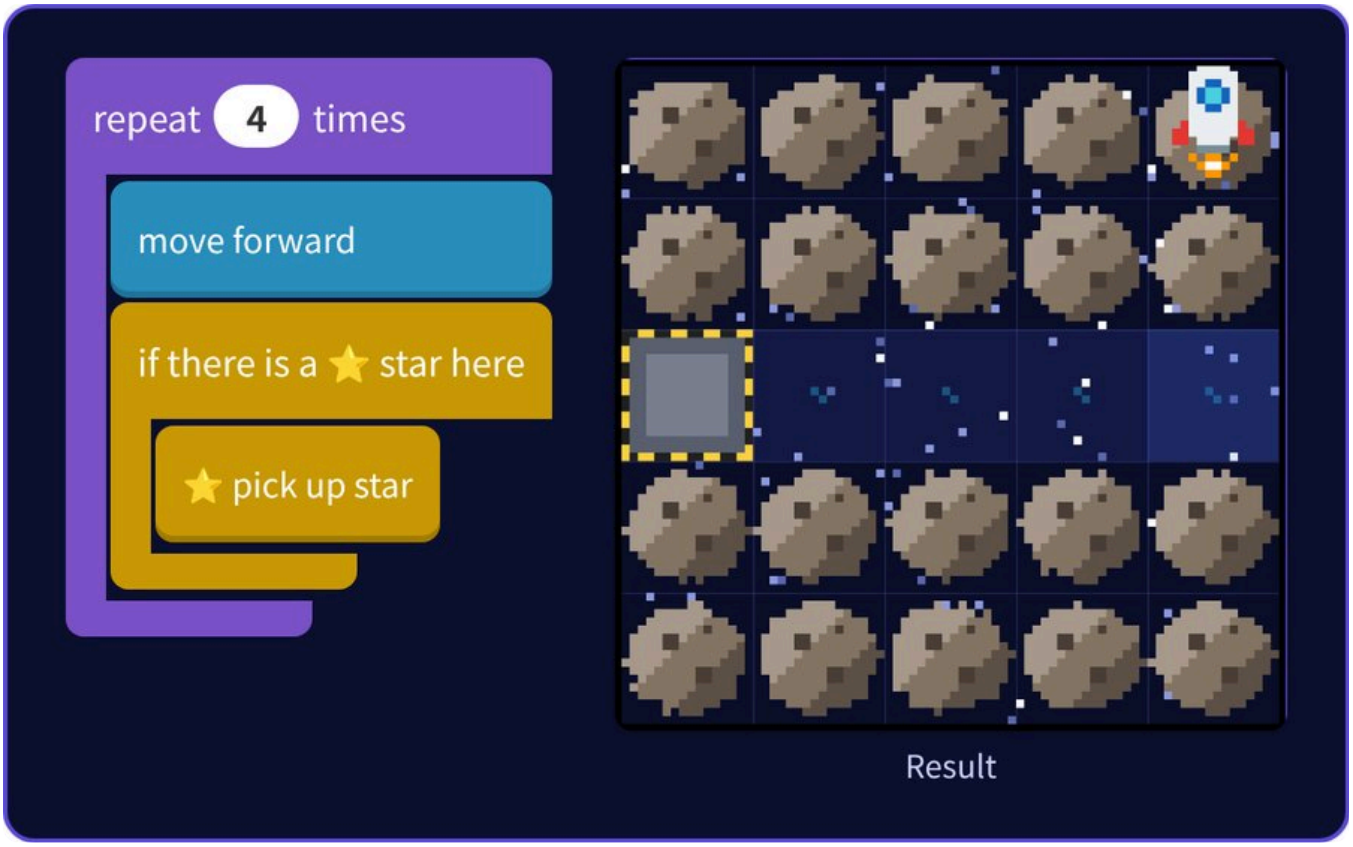
💡 **Hit (look if you get stuck)**

"repeat 4: move forward → if there is a star here, pick it up"

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 5: If there is a star




✨ **Key point**

Pick up only when there is a star. Check with “if” before you act.

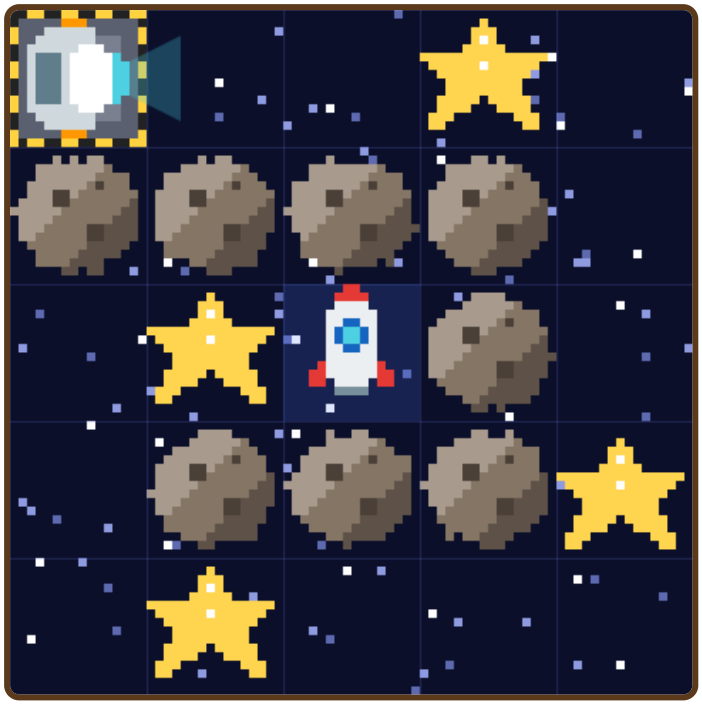
⚠ **Common mistake**

Picking up every time without “if” fails on empty squares.

🎬 Video: `videos/en/space-05.mp4`

# Stage 6: Until mission complete

Conditions | Repeat until



The start

**Problem**

Stars are scattered around a spiral. Use "repeat until mission complete" and check as you go.

🎯 Target: 6 blocks or fewer

🎯 **Goal**：Repeat until the mission is complete

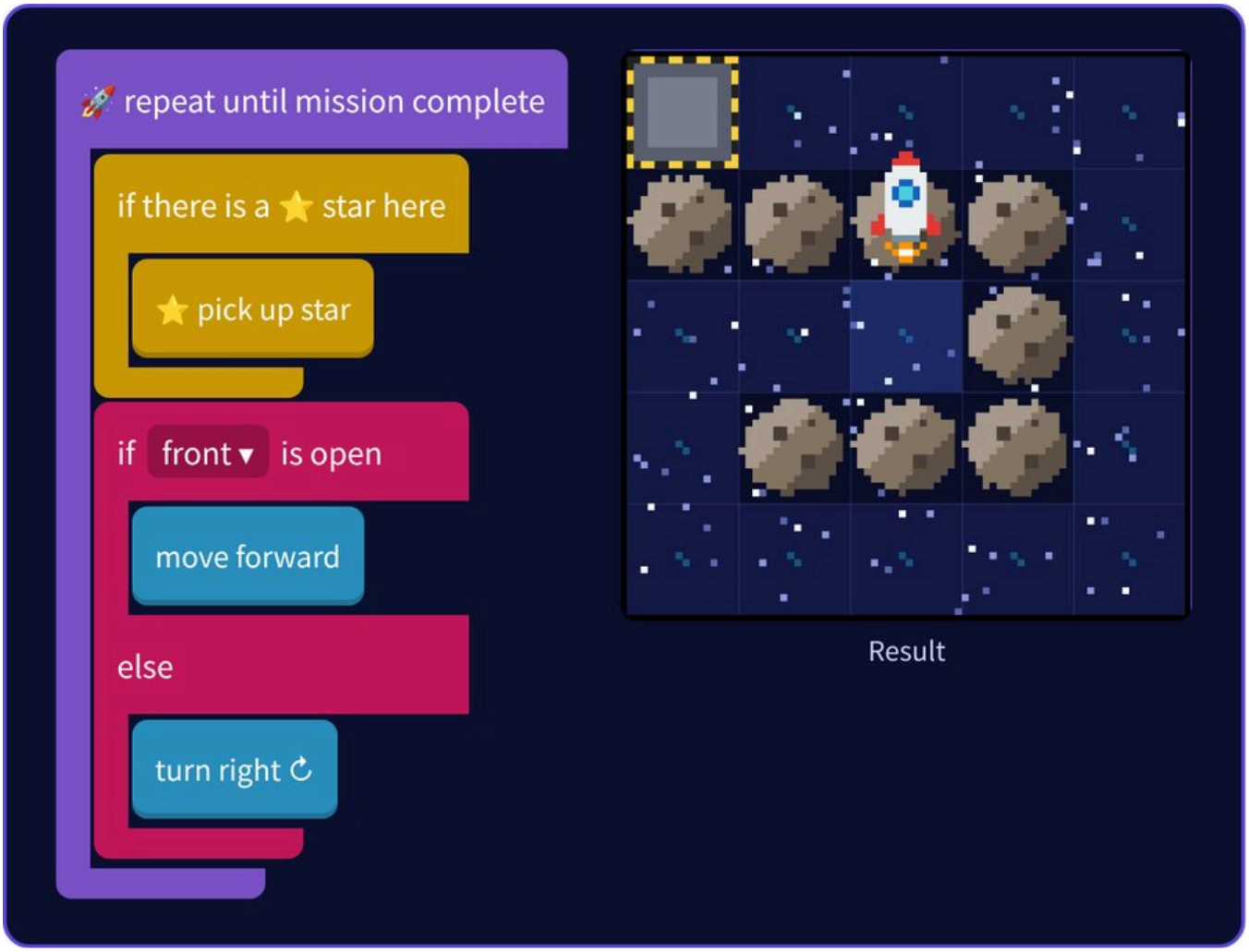
💡 **Hint (look if you get stuck)**

Inside the loop: if there is a star, pick it up → if front is open, move forward, else turn right

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 6: Until mission complete




✨ **Key point**

Each time, first check under your feet, then move.

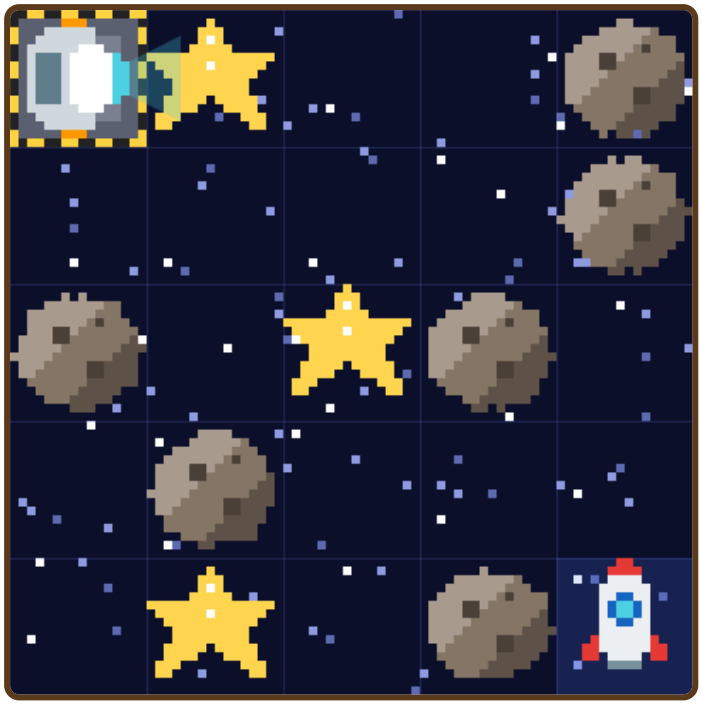
⚠ **Common mistake**

Forgetting to check at the corners leaves stars behind.

🎬 Video: `videos/en/space-06.mp4`

# Stage 7: Any space!

Generalization | Conditions | Repeat until



The start

Example: the world changes every time you run

**Problem**

Final challenge! Your program is tested in 30 random spaces. Collect all the stars and board the rocket to clear!

🎯 **Goal**：Clear any space

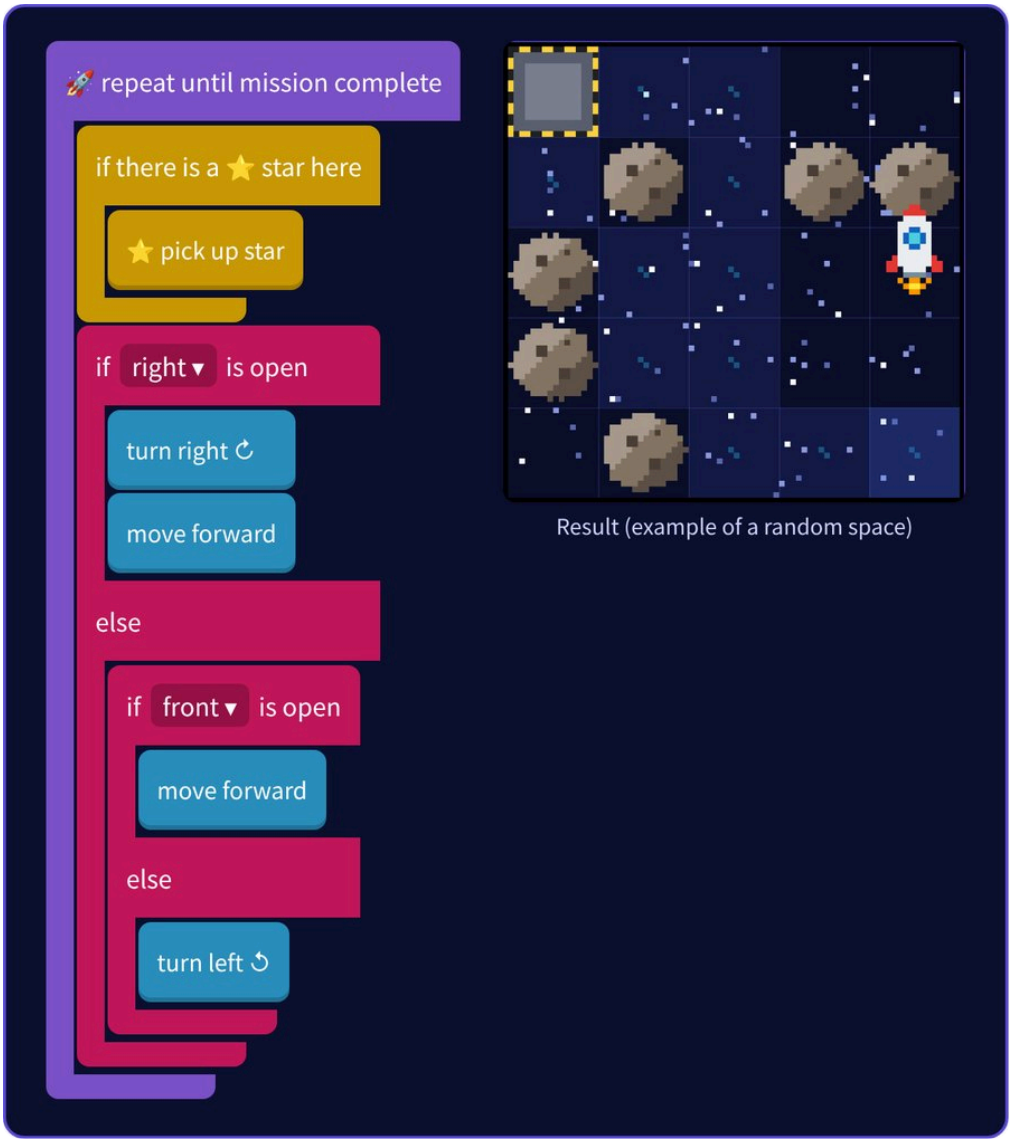
💡 **Hint (look if you get stuck)**

Use the "right-hand rule" to follow the walls, and pick up stars when you find them. It's OK to pass the rocket — keep walking until you have every star.

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer  Stage 7: Any space!




Result (example of a random space)

> ✨ **Key point**
>
> Right-hand rule + pick up stars. It is OK to pass the rocket!

> ⚠ **Common mistake**
>
> You may think it stops at the rocket — it keeps going until every star is collected.

🎬 Video: `videos/en/space-07.mp4`

Column

## Check first, then act

In stage 5, some squares have stars and some don’t. If you memorize “pick up on squares 2 and 4”, it breaks when the stars move. If you write “if there is a star, pick it up”, it works for any layout. That is the power of conditions.

Chapter 3

# 🐍 Space in Python

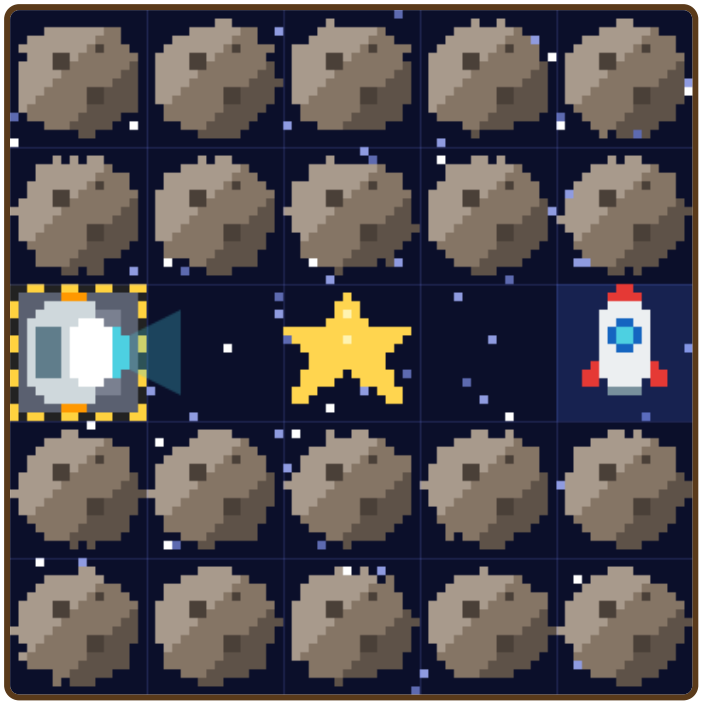
Write star hunting in **Python**: `forward()`, `turn_right()`, `pick_up()`; check with `if has_star():`, and repeat to the end with `while not mission_done():`.

**Commands in this chapter**

forward() / turn_right() / turn_left() / pick_up() / for i in range(4): / if has_star(): / while not mission_done():

# Stage 1: Pick up a star

Sequence | Python syntax (colon & indent)



The start

**Problem**

Pick up the star ⭐ with pick_up(), then go to the rocket 🚀. pick_up() works only on a star square.

🎯 **Goal**：Use pick_up()

💡 **Hint (look if you get stuck)**

forward() ×2 → pick_up() → forward() ×2

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 1: Pick up a star

```
forward()
forward()
pick_up()
forward()
forward()
```

**✨ Key point**

pick_up() picks up the star on your square. Collect them all, then go to the rocket!

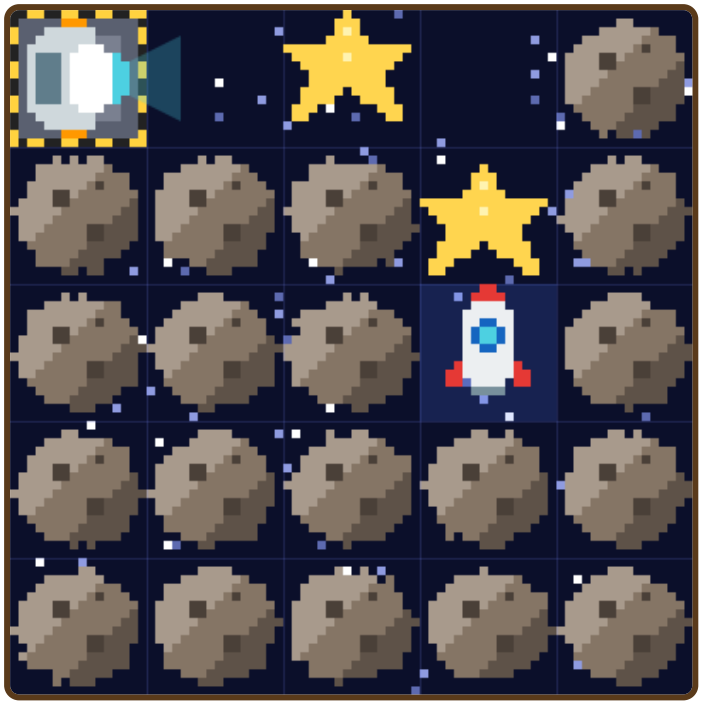
**⚠ Common mistake**

Writing pickup() or pick up() causes a NameError. Don’t forget the “_”.

🎬 Video: `videos/en/pyspace-01.mp4`

# Stage 2: Turn and collect

Sequence | Direction | Python syntax (colon & indent)



The start

**Problem**

There are 2 stars. Use turn_right() where the path turns.

🎯 **Goal**：Turn and collect in order

💡 **Hint (look if you get stuck)**

forward() ×2, pick_up() → forward() → turn_right() → forward(), pick_up() → forward()

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 2: Turn and collect

```
forward()
forward()
pick_up()
forward()
turn_right()
forward()
pick_up()
forward()
```

**✨ Key point**

turn_right() only changes direction. The order of turning and picking up matters.

**⚠ Common mistake**

pick_up() on an empty square fails. Check which square has the star.

🎬 Video: `videos/en/pyspace-02.mp4`

# Stage 3: Collect with for

Loops | Python syntax (colon & indent)



The start

**Problem**

The stars are in a row. Repeat “forward() → pick_up()” with for. Indent the lines inside the loop by 4 spaces.

🎯 **Goal**：Repeat with for

💡 **Hint (look if you get stuck)**

for i in range(3): → (indented) forward() and pick_up() → forward()

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 3: Collect with for

```
for i in range(3):
    forward()
    pick_up()
forward()
```

**✨ Key point**

To put 2 lines inside for, indent both by 4 spaces. The last forward() is not indented.

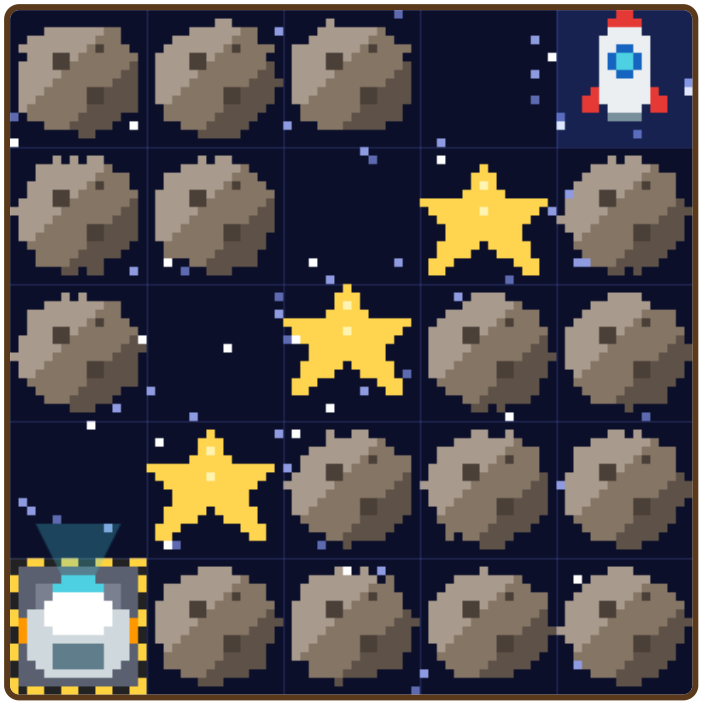
**⚠ Common mistake**

If you forget to indent pick_up(), it runs only once.

🎬 Video: `videos/en/pyspace-03.mp4`

# Stage 4: Star stairs

Loops | Patterns | Python syntax (colon & indent)



The start

**Problem**

Find the set of moves that repeats on the stairs and put all of it inside the for loop.

🎯 **Goal**：Find the set that repeats

💡 **Hint (look if you get stuck)**

for i in range(4): forward(), turn_right(), forward(), pick_up(), turn_left()

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 4: Star stairs

```
for i in range(4):
    forward()
    turn_right()
    forward()
    pick_up()
    turn_left()
```

**✨ Key point**

Put the whole 5-line set inside for. You reach the rocket during the 4th repeat, and it ends there.

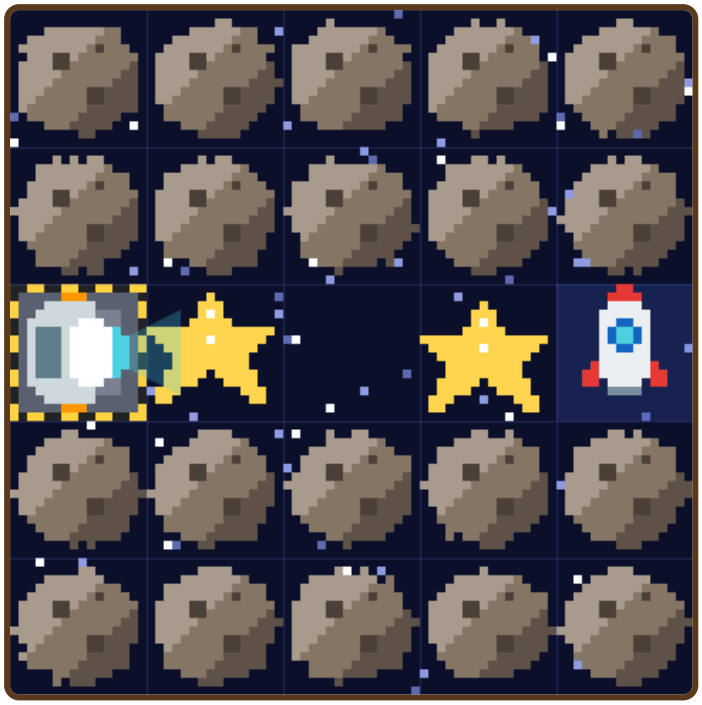
**⚠ Common mistake**

Forgetting turn_left() keeps changing your direction and you hit an asteroid.

🎬 Video: `videos/en/pyspace-04.mp4`

# Stage 5: if has_star():

Conditions | Loops | Nesting | Python syntax (colon & indent)



The start

**Problem**

Some squares have stars and some don’t. has_star() checks the square you are on. Use if so you only pick up where there is a star.

🎯 **Goal**：Check with if before you act

💡 **Hint (look if you get stuck)**

for i in range(4): forward() → if has_star(): → (indent more) pick_up()

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer　Stage 5: if has_star():

```
for i in range(4):
    forward()
    if has_star():
        pick_up()
```

✨ **Key point**

Lines inside if are indented one more level (8 spaces in total).

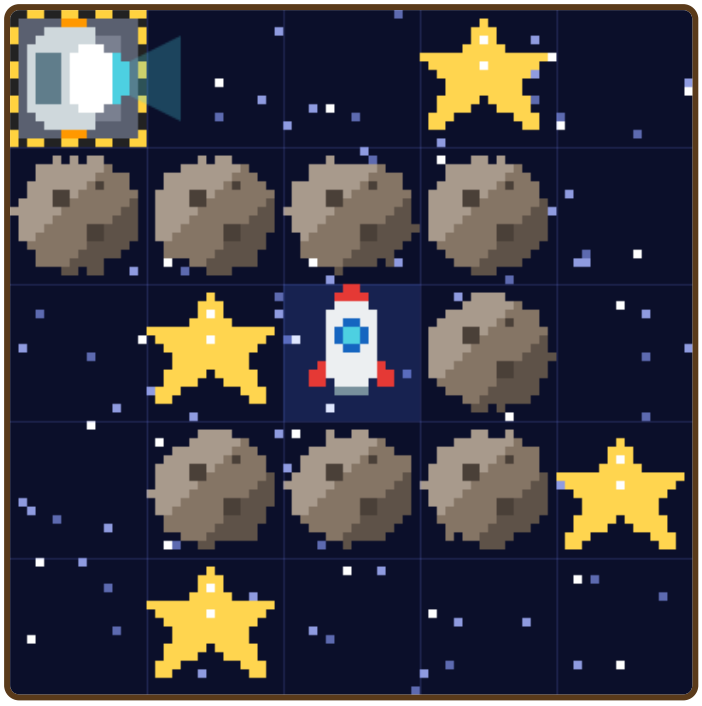
⚠ **Common mistake**

Forgetting the “:” after if has_star() causes a SyntaxError.

🎬 Video: `videos/en/pyspace-05.mp4`

# Stage 6: while not mission_done():

Conditions | Repeat until | Nesting | Python syntax (colon & indent)



The start

**Problem**

Stars are scattered around a spiral. Repeat with “while not mission_done():” and check as you go.

🎯 **Goal**：Repeat with while not mission_done():

💡 **Hint (look if you get stuck)**

Inside while: if has_star(): pick_up() → if is_open('front'): forward() else: turn_right()

✏ **Write your own idea**

The answer is on the next page →

# 📖 Answer  Stage 6: while not mission_done():

```
while not mission_done():
    if has_star():
        pick_up()
    if is_open('front'):
        forward()
    else:
        turn_right()
```

**✨ Key point**

mission_done() becomes True when you have every star and are at the rocket. not repeats “while it is not done yet”.

**⚠ Common mistake**

If you join the two ifs with elif, you will not move on the turn you pick up a star (it can still clear).

🎬 Video: `videos/en/pyspace-06.mp4`

Column

## How to read while not

`while not mission_done():` means “repeat while the mission is not done”. `not` means “is not”. Indent the lines inside by 4 spaces.

# Well done!

You worked hard to the end! Let’s look back at the ideas you used.

| Idea | Where did it appear? |
|---|---|
| Order | Line up move, turn and pick up (stages 1–2) |
| Loops & patterns | Pack “move → pick up” together (stages 3–4) |
| If (conditions) | Pick up only when there is a star (stage 5) |
| Repeat until | Until the mission is complete (stages 6–7) |
| Python | Write the same thing in words (code) |

When you clear every stage, get your certificate from “⚙ Settings” → “📜 Certificate”.

PART 2

# Part 2 Guide for teachers and parents

# About this guide

This book covers “Space Star Hunt” (7 stages) and “Python Space” (6 stages). Part 1 is written in simple language for students.

The core idea is a condition: “check the situation, then act”. Picking up where there is no star fails, which makes conditions necessary. The final stage is tested on 30 random worlds, so students need a general method.

## Aims

- Combine sequence, loops and conditions to reach a goal.
- Notice and explain the value of checking before acting.
- In Python, learn how to write if, while and function calls.

# Overview of the stages

| App | # | Stage | Ideas |
|---|---|---|---|
| 🚀 Space Star Hunt | 1 | Pick up a star | Sequence |
| | 2 | Turn and collect | Sequence, Direction |
| | 3 | Collect with a loop | Loops |
| | 4 | Star stairs | Loops, Patterns |
| | 5 | If there is a star | Conditions, Loops |
| | 6 | Until mission complete | Conditions, Repeat until |
| | 7 | Any space! | Generalization, Conditions, Repeat until |
| 🐍 Python Space | 1 | Pick up a star | Sequence, Python syntax (colon & indent) |
| | 2 | Turn and collect | Sequence, Direction, Python syntax (colon & indent) |
| | 3 | Collect with for | Loops, Python syntax (colon & indent) |
| | 4 | Star stairs | Loops, Patterns, Python syntax (colon & indent) |
| | 5 | if has_star(): | Conditions, Loops, Nesting, Python syntax (colon & indent) |
| | 6 | while not mission_done(): | Conditions, Repeat until, Nesting, Python syntax (colon & indent) |

# Suggested order and teaching points

| Students | Stages | Lessons | Key ideas |
|---|---|---|---|
| Grades 3–4 | 1–4 | 1–2 | Order, loops, patterns |
| Grades 4–6 | 5–7 | 2 | Conditions, repeat until, generalization |
| Grade 6 – junior high | Python 1–6 | 2–3 | if/while, function calls |

## Common difficulties and support

- **Picking up where there is no star:** in stage 5, show that a memorized route breaks when the layout changes, and draw out the idea of “checking”.
- **Nesting a condition in a loop:** have students say one square’s action in words (move → if star, pick up) before building.
- **Finding the pattern (stage 4):** trace the stairs and circle the repeating “set”.
- **Python:** confirm what `while not` means and that indentation decides what is inside.

## Example 45-minute lesson (stage 5)

| Time | Activity | Support |
|---|---|---|
| Intro 5 min | Notice that some squares have stars and some don’t; share the goal. | Recall “move → pick up”. |
| Main 30 min | Build a route for this layout; see it break when the layout changes; rebuild with “if”. | Record the star count on Worksheet 5. |
| Wrap-up 10 min | Summarize the value of conditions; talk about real-life examples. | Automatic doors, sensor lights. |

# Assessment (example)

| Area | Meets expectations (B) | Exceeds (A), e.g. |
|---|---|---|
| Knowledge & skills | Combines blocks (code) correctly and completes the task. | Uses loops and conditions to complete it with a short, clear program. |
| Thinking, judgment & expression | Finds where the error is from the result and fixes it. | Explains the rule or steps in words and makes a program that works in other cases too. |
| Attitude toward learning | Keeps trying even when it doesn’t work. | Shares ideas with friends and looks for better ways on their own. |

Use the worksheets (1 reflection; 2, 5, 7 etc.) and the app progress (8 check card, certificate) as evidence.

# Using the apps and related materials

## Requirements and tips

- Works in the browser on computers, tablets and smartphones. No installation needed. An internet connection is required.
- Cleared stages are saved in that device’s browser. If devices are shared, check progress at the start of each lesson.
- **Teacher mode:** add `?sensei` to the end of the address to show an “Answer” button.
- From “⚙ Settings” you can switch Japanese/English, use the pixel font, switch to the Python version and show the certificate.

## Related materials

| Material | Contents |
|---|---|
| Guide (guide.html) | Goals, common mistakes, hints and answers for every task. This book is built from the same data. |
| Worksheets (PDF) | 5 Space / Break & Build (order and count table), 7 Python, 1 goal & reflection, 8 check card. |
| Lesson plans (Japanese) | Example 45-minute lessons (open them from each task in the Guide). |
| Videos (videos.html) | Short videos showing how to solve each task. |
| Lesson slides | 03 Space Star Hunt, 07 Python Space (PowerPoint). |